

B.TECH PROJECT
AI PROBLEM SCOPING

Predicting University Enrollment Yield

A structured 8-step scoping framework for translating a business request into a measurable, ethical, machine-learning problem.

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STEP
—
AGENDA

Eight steps from business problem to AI solution

01 Understanding the Business Problem

02 Defining the Problem Precisely

03 Is AI Really Needed?

04 Identifying Stakeholders

05 Defining the AI Task

06 Defining the Success Metrics

07 Operational and Business Constraints

08 Assumptions and Risks

STEP
1

UNDERSTANDING THE BUSINESS PROBLEM

Suppose a request came from university side , and they say -

“

We don't know how many students are going to join our college this year. We need an AI to predict admissions.

ADMISSIONS

Manual 25% rule is drifting — fell to 21% last year.

FINANCE & OPS

Under-enrollment = lost tuition; faculty payroll at risk.

HOUSING

Over-enrollment = expensive hotel overflow; hard bed cap.



**STEP
2****DEFINING THE PROBLEM PRECISELY****From Poor problem statement to a measurable target****POOR PROBLEM STATEMENT**

"Predict the total number of students joining the college next year."

PRECISE PROBLEM STATEMENT

Predict the individual enrollment probability (Yield) for every admitted student based on engagement and financial profile, aggregating these probabilities to forecast total incoming class size with a margin of error < 3%, finalised 3 months before the semester starts.

WHO?

Admitted
students

WHAT?

P(accept offer &
pay deposit)

WHEN?

Continuous; final
before 3 month

WHY?

Plan finance,
housing, academics

OUTCOME?

No budget deficits
or housing crises

STEP
3

IS AI REALLY NEEDED?

Why the historical-average rule breaks and ML needed

RULE-BASED APPROACH

Total Expected = Offers × 0.25

Why it fails

- This rule-based approach is fundamentally flawed because it treats every single student identically. It assumes Applicant A and Applicant B both have exactly a 25% chance of joining
- Breaks when a competitor cuts tuition.
- Yield already drifted: 25% → 21% in one year.

MACHINE LEARNING APPROACH

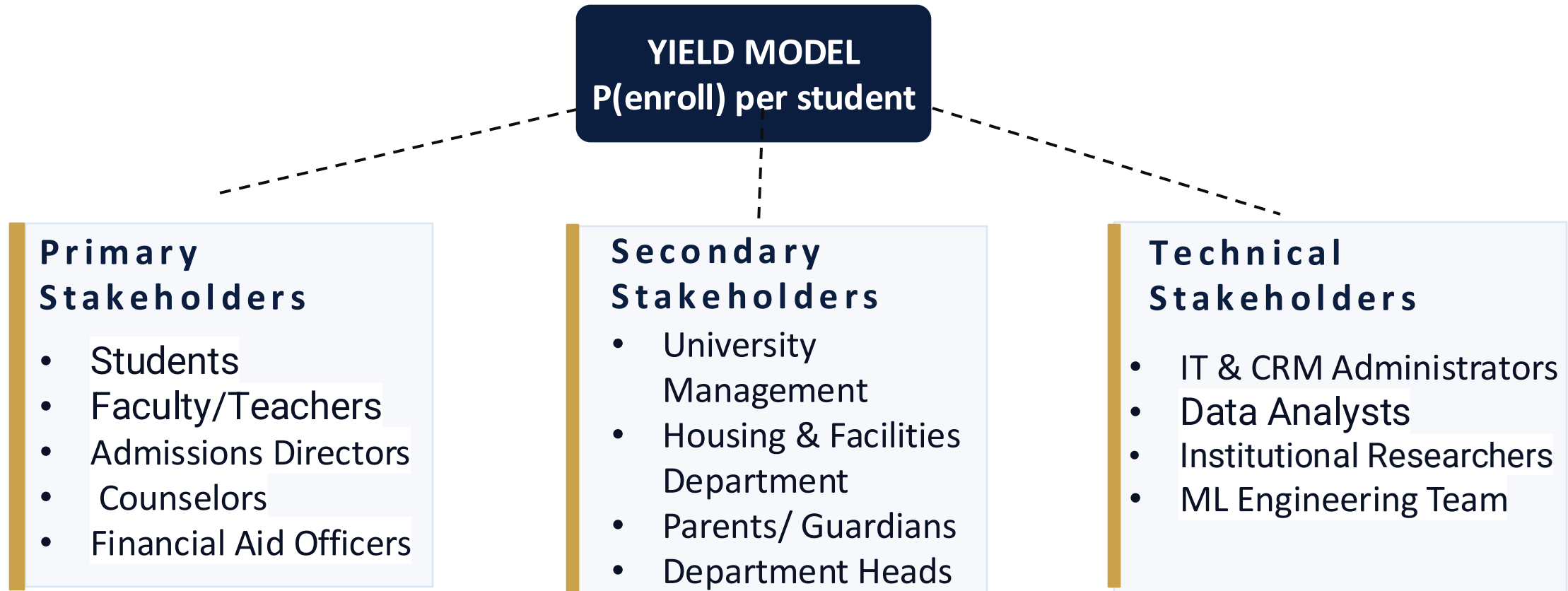
*Weighs many signals per student, for 10,000+ applicants — simultaneously.
Works on Complex Patterns*

Signals the model learns from

- Distance from hometown to campus
- Campus tour attendance (strong signal)
- Speed of opening the admission email
- Scholarship / financial aid amount
- Competitiveness of the chosen major

**STEP
4****IDENTIFYING STAKEHOLDERS**

That has an interest or affected by the actions, decisions and outcomes of a project



**STEP
5****DEFINING THE AI TASK**

Translate a broad business objective into a mathematical and computational problem that an algorithm can actually solve.

**STUDENT A · HIGH LIKELIHOOD****0.85**

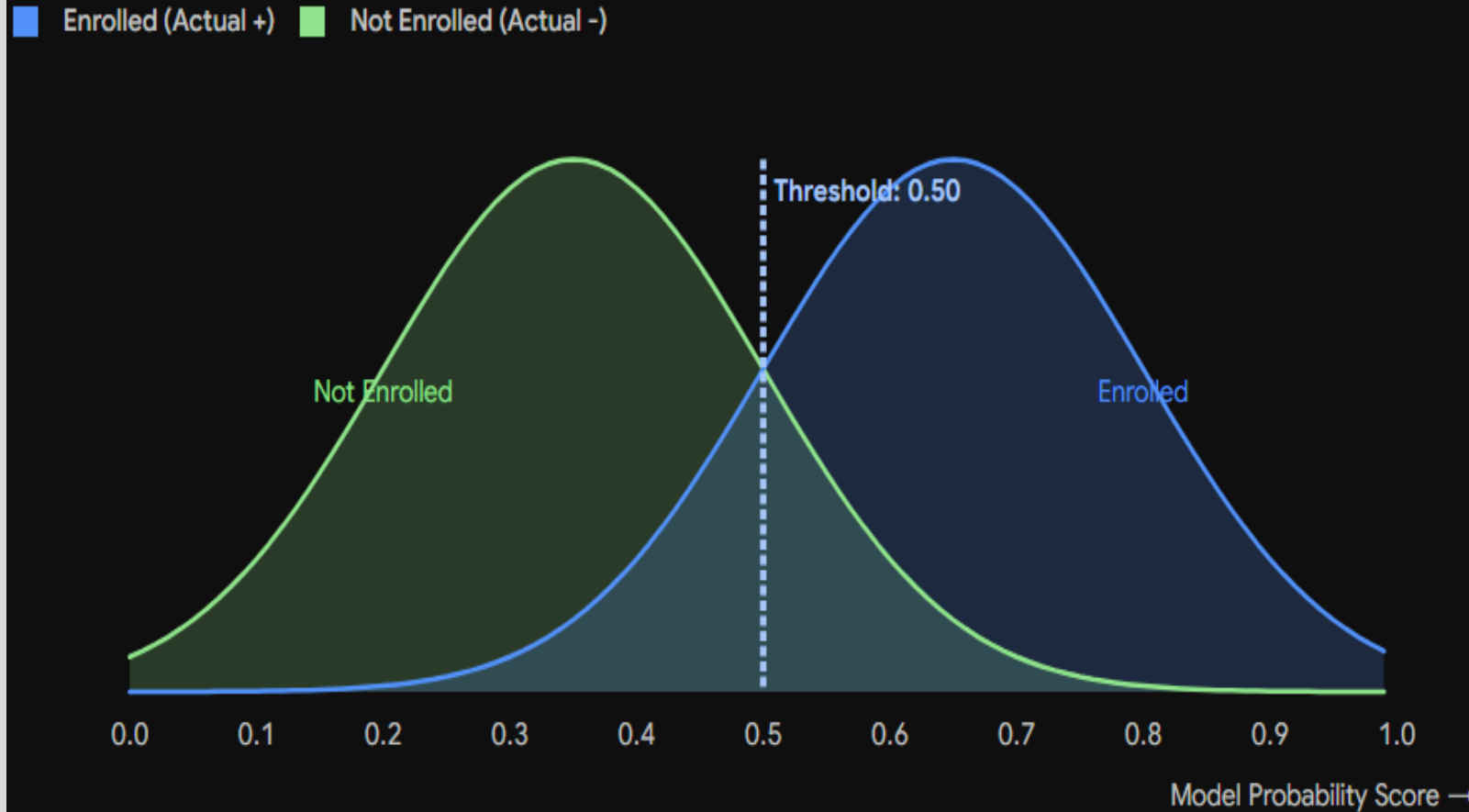
Attended campus tour, opened email within 1 hour, lives 40 km away, received 30% scholarship.

STUDENT B · LOW LIKELIHOOD**0.12**

No campus visit, slow email engagement, lives in another country, minimal aid offered.

See the Trade-off in Action

Admissions AI Model Simulator



Adjusting the "Threshold" (the probability cut-off where the AI decides a student is a "Yes" instead of a "No") drastically changes your Precision and Recall.

F1 Score (The Balancing Act)
The harmonic mean of Precision and Recall. It forces to find a balance between the two.

ROC AUC (The Ranking Power)
Area Under the Receiver Operating Characteristic Curve. It measures the model's ability to distinguish between classes.

STEP
6

DEFINING THE SUCCESS METRICS

Success metrics has two categories: **Technical Metrics** (to prove the math works) and **Business Metrics** (to prove the university is saving money).

ROC-AUC (Target: > 0.80): This is primary metric for the individual classification. It proves model can accurately rank students. ROC-AUC ensures the actual enrolled students are at the very top.

Total Headcount Variance (Target: < 3% Error): If model predicts 2,000 students will join, and 2,040 actually join, the variance is 2%. This means the CFO can confidently set the university budget without fear of a sudden deficit.

Metric / Component	Value	Description
Accuracy	88.8%	Overall correct rate
Precision	83.6%	If model says 'Enroll', how often is it right?
True Positive	1991	Enrolled & Predicted Enroll

Metric / Component	Value	Description
False Positive	118	Didn't Enroll but Predicted Enroll
True Negative	632	Didn't Enroll & Predicted No
False Negative	44	Enrolled but Predicted No

**STEP
7****OPERATIONAL AND BUSINESS CONSTRAINTS**

Every AI Model cooperates under some constraints

OPERATIONAL

Must integrate into the existing CRM

Counselors see a "Likelihood to Enroll" score next to each student name — not on a separate dashboard.

DATA · THE BLIND SPOT

Competitor offers are invisible

The model only knows what the university knows. It cannot see information may like mental health indicators , family circumstance, updated financial records

ETHICAL · CRITICAL

No admissions redlining

If the model is used to *deny* admission to a student simply because the AI thinks they won't enroll (perhaps due to their financial background), that is highly unethical and potentially illegal.

STEP
8

ASSUMPTIONS AND RISKS

What the model trusts — and what could invalidate it

ASSUMPTIONS

What we take for granted

- Past 3–5 years of enrollment data reliably reflects future student behaviour.
- University tuition and reputation stay broadly stable relative to competitors.
- We assume that a "Portal Login" or an "Email Open" event recorded in the database was generated by the *actual student* showing intent, rather than a web-crawler

RISKS

What could break the model

- Macro shocks — recession, pandemic, loan-policy changes — invalidate historical patterns.
-
- Competitor actions — a students often pay a ₹25,000 "Seat Booking Deposit" in May just as a backup. The AI marks Enrolled = True. In late July, a state government counseling list clears , the student forfeits deposit and vanishes 4 days before orientation

Industry Adoption & SaaS Solutions

Why private institutions are turning to Machine Learning to predict individual student behaviour?

The Admission Divide:

- *Government Institutes (IITs/NITs):* Centralized, rank-based algorithms (JoSAA) with 100% yield. No ML required.
- *Private Universities (SRM, Ashoka, BITS):* High competition, overlapping offers, and unpredictable student behavior.

The Operational Bottleneck:

- Institutions process 50,000+ inquiries annually.
- Manual tracking and historical average guessing leads to severe budget deficits and housing misallocations.

The Market Leaders (B2B EdTech companies):

- **Meritto (formerly NoPaperForms):** Used by 1,000+ institutions. Utilizes predictive analytics and agentic AI ("Mio") to forecast total enrollments and flag drop-off risks.
- **ExtraaEdge:** Deploys "AI Lead Intent Scoring" using behavioral funnel signals to prioritize counselor follow-ups

Core ML Capabilities in Production:

- **Dynamic Resource Allocation:** Directing human counselors to call only the highest-converting leads.
- **Lead Intent Scoring:** Assigning a 1–100 probability score to every applicant.

Thank you.

*Questions on scoping, modeling choices, or risks —
happy to discuss.*

